

Telco Customer Churn: Analysis Guide & Product Insights

Project Goal: Understand churn drivers and frame retention strategies for product/operations teams

QUICK CONTEXT

Why This Matters: In telecom, acquiring a new customer costs 5–7x more than retaining an existing one. A 5% improvement in churn retention can increase profitability by 25–30%. This analysis identifies *where* and *why* customers leave, so you can:

- Target retention budgets efficiently
 - Design better onboarding
 - Build bundled products that stick
 - Optimize pricing strategy
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ANALYSIS STRUCTURE & KEY FINDINGS

SECTION 1: Baseline Metrics

What You're Looking For:

- **Overall Churn Rate:** Should be 20–27% for typical telecom (benchmark)
- **Avg Tenure - Churned vs. Retained:** Churned customers should have <12 months tenure on average
- **Revenue Lost:** \$ or % of total revenue that churned customers represent

Sample Output:

Total Customers: 7,043
Churned: 1,869 (26.5%)
Revenue Lost: \$3.7M (lifetime value)

Churned customers avg tenure: 17 months
Retained customers avg tenure: 37 months

Business Narrative:

"Our churn rate of 26.5% is in the competitive range, but churned customers exit at just 17 months—much earlier than retained customers (37 months). This indicates an **early-stage problem**: we're losing customers during the critical first 1–2 years. Focus: improve onboarding and 6-month experience."

SECTION 2: Contract Type 🔥

What You're Looking For:

- Month-to-month churn should be **3x higher** than 2-year
- 2-year contracts should have <15% churn (lock-in effect)

Sample Output:

Contract Type	Customers	Churn Rate	Avg Tenure	Avg Monthly Charge
Month-to-month	3,875	42.7%	17 mo	\$65.33
One year	1,473	11.3%	29 mo	\$61.08
Two year	1,695	2.8%	48 mo	\$59.61

Business Narrative:

"Contract type is the **single strongest predictor** of churn. Month-to-month customers churn at 42.7%—nearly **15x** the rate of 2-year customers (2.8%). This is the #1 lever: moving customers from month-to-month to 1-year or 2-year contracts directly reduces churn risk. Implication: offer 6–12 month incentives (free premium channels, security suite discount) to upgrade contracts."

Interview Talking Point:

"The data shows contract flexibility creates switching risk. My recommendation: A/B test a 'contract upgrade offer' targeting month-to-month customers with >\$50 tenure. If we move even 15% of MTM to 1-year, churn drops by 5–7 percentage points."

SECTION 3: Price Sensitivity 💰

What You're Looking For:

- Budget customers (<\$25/mo) should churn <15%
- Premium customers (>\$75/mo) may have higher churn (value perception gap)
- Sweet spot: \$50–75/mo

Sample Output:

Price Segment	Customers	Churn Rate	Avg Monthly
Budget (<\$25)	1,200	12.4%	\$20.50
Low-Mid (\$25-50)	2,100	18.7%	\$39.25
Mid-High (\$50-75)	2,300	29.1%	\$63.88
Premium (>\$75)	1,443	35.2%	\$87.50

Business Narrative:

"Price sensitivity becomes critical above \$50/month. Customers paying >\$75/month churn at 35.2%—nearly **3x** the rate of budget customers. This suggests two problems: (1) Premium customers feel overpriced relative to alternatives, OR (2) They signed up for a specific feature (fiber, premium support) that isn't delivering value. Strategy: audit premium-tier satisfaction and ensure value delivery. Consider tiered pricing with value-adds (e.g., free premium streaming for >\$75/mo)."

SECTION 4: Service Adoption & Stickiness

What You're Looking For:

- 0 services (internet only) should churn 25–35%
- 3+ services should churn <10%
- Tech Support & Security adoption should correlate with lower churn

Sample Output:

Service Adoption	Customers	Churn Rate	Avg Monthly
0 Services	3,498	26.5%	\$40.12
1 Service	1,750	16.2%	\$58.37
2 Services	1,200	9.1%	\$71.44
3 Services	595	3.4%	\$85.22

Business Narrative:

"**Multi-product adoption is the strongest retention mechanism.** Customers with 3+ add-ons (security, tech support, backup) churn at just 3.4%—nearly **8x lower** than internet-only. This is habit formation: more products = higher switching cost = stickier customer. **Cross-sell strategy:** Implement early nudges (30-day/60-day post-signup) to trial security or tech support at discounted rates. Measure: if we move 20% of internet-only to 2+ services, churn drops 3–4%."

Individual Service Impact:

Tech Support:

- Yes: 14% churn
- No: 31% churn

(Reduction: 17 percentage points!)

Online Security:

- Yes: 12% churn
- No: 28% churn

(Reduction: 16 percentage points!)

Online Backup:

- Yes: 15% churn
- No: 27% churn

(Reduction: 12 percentage points!)

Insight: Tech Support and Security are "hero" products—offer them bundled at first signup.

SECTION 5: Tenure & Onboarding ⚠️

What You're Looking For:

- Months 0–6 should show the highest churn drop-off
- By month 12, churn rate should stabilize

Sample Output:

Tenure Cohort	Customers	Churn Rate
0-6 months	742	54.3% ← CRISIS POINT
7-12 months	823	31.2%
13-24 months	1,245	14.5%
25-36 months	1,150	8.3%
37-60 months	1,650	4.1%
60+ months	1,393	3.7%

Business Narrative:

"Our **onboarding is broken**. Customers aged 0–6 months churn at 54.3%—nearly **15x** the rate of 60+ month customers. This is our biggest leak. By month 13, churn stabilizes at 14.5%, suggesting: (1) Poor first-month experience, (2) Unclear value prop, or (3) High setup friction. **Immediate action:** Conduct exit interviews with 0–6 month churners. Most likely: service quality issues, installation problems, or unclear billing. Hypothesis: proactive support calls at 2 weeks + free month 2 = 10–15% churn reduction."

High-Risk New Customer Segment (Month-to-Month + 0–6 Months):

New MTM Customers (Tenure $\leq$ 6 mo):

- Segment Size: 487
- Churn Rate: 68.2%
- Revenue at Risk: \$380K

Insight: This is your "save now" cohort. Every dollar spent on retention here has 5–10x ROI.

SECTION 6: Internet Service Quality INFRASTRUCTURE MATTERS

What You're Looking For:

- Fiber churn should be 15–20%
- DSL churn should be 25–35%
- No internet = highest churn (likely very old product)

Sample Output:

Internet Service	Customers	Churn Rate	Avg Tenure	Avg Monthly
Fiber optic	3,096	11.6%	42 mo	\$84.30
DSL	2,400	25.8%	28 mo	\$47.50
No internet	1,547	45.2%	18 mo	\$35.40

Business Narrative:

"Infrastructure quality directly drives loyalty. Fiber customers churn at 11.6% vs. DSL at 25.8%—a **2.2x difference**. This is product differentiation: fiber = faster speeds = better user experience = stickier. DSL customers are 4 times more likely to churn than fiber. **Implication:** (1) Prioritize fiber expansion to DSL areas, (2) Offer free/discounted fiber upgrades to high-value DSL customers. **ROI calculation:** If we convert 30% of DSL to fiber and reduce their churn by 10%, we save ~120 customers/month = \$600K annual revenue."

Contract × Internet Interaction:

Fiber optic + 2-year: 3.2% churn
Fiber optic + Month-to-month: 18.5% churn

DSL + 2-year: 12.1% churn
DSL + Month-to-month: 37.4% churn

Insight: Even fiber can't overcome month-to-month flexibility. Combined effect: both matter.

SECTION 7: Demographics 👤

What You're Looking For:

- Senior citizens (65+) should have higher churn (price or service sensitivity)
- Identify if support needs are being met

Sample Output:

Age Group	Customers	Churn Rate	Avg Monthly
Senior	1,142	41.7%	\$61.25
Non-Senior	5,901	22.8%	\$62.88

Business Narrative:

"Senior citizens churn at nearly **2x** the rate of non-seniors (41.7% vs. 22.8%). **However**, they pay similar monthly charges, suggesting the problem isn't price but **service satisfaction or usability**. Hypothesis: seniors lack tech support access (difficulty online) or need simpler plans. **Action**: Create senior-friendly bundles with included phone support (not email-based). Segment and monitor: dedicated support team or support hotline (not chat-based)."

SECTION 8: Payment Method 💳

What You're Looking For:

- Autopay (credit card, automatic bank transfer) = lower churn
- Manual payment (checks, money order) = higher churn (disengagement signal)

Sample Output:

Payment Method	Customers	Churn Rate
Mailed Check	1,613	45.3%
Electronic Check	1,567	35.7%
Credit Card (autopay)	2,100	16.4%
Bank Transfer (autopay)	1,763	12.8%

Business Narrative:

"**Payment friction is a leading indicator of churn**. Check-paying customers churn at 45.3% vs. autopay at 12.8%—a **3.5x difference**. This isn't correlation; it's causation: manual payment → easy to forget/cancel → higher churn. **Immediate action**: Incentivize autopay enrollment (e.g., \$5/month discount). Secondary insight: electronic check (35.7%) is middle ground—not as sticky as credit card. **Target**: Convert top 20% of check-payers to autopay = 8–10% churn reduction in that cohort."

SECTION 9: High-Risk Segment Scoring

Purpose: Identify which customers need proactive retention outreach *now*.

Sample Output:

Risk Tier	Segment Size	Churn Rate	Retained Revenue	Lost Revenue
CRITICAL	234	71.4%	\$3.2K/mo	\$8.1K/mo
HIGH	687	64.2%	\$12.4K/mo	\$21.3K/mo
MEDIUM-HIGH	1,240	48.3%	\$45.2K/mo	\$41.6K/mo
MEDIUM	2,150	23.5%	\$98.7K/mo	\$30.2K/mo
LOW	2,732	4.1%	\$145.3K/mo	\$6.1K/mo

Risk Definitions:

- **CRITICAL:** Month-to-month + <12 mo tenure + >\$65/mo charges
- **HIGH:** Month-to-month + <6 mo tenure
- **MEDIUM-HIGH:** Month-to-month + <12 mo tenure
- **MEDIUM:** <6 mo tenure (any contract)
- **LOW:** Everything else

Business Narrative:

"We have **234 CRITICAL-risk customers** who are losing us ~\$8.1K/month. These customers have zero lock-in, are new (<12 mo), and pay premium rates. They're the most likely to churn AND have the highest value.

Immediate action: Assign retention team to call all CRITICAL customers within 2 weeks. Offer: 50% discount on tech support + security bundle if they upgrade to 1-year contract. **ROI:** Saving 50% of CRITICAL and 25% of HIGH = \$14.7K/month retention = \$176K annually."

Revenue Impact Summary:

CRITICAL + HIGH combined:

- Segment Size: 921 customers
 - Current Monthly Revenue at Risk: \$29.4K
 - If we save 40% via intervention: \$11.8K/month = \$142K/year
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SECTION 10: Window Functions

What You're Showing: Advanced ranking and cumulative analysis

Example Output — Cumulative Churn by Tenure:

Tenure	Customers	Churned	Churn%	Cumulative Churned	Cumulative Customers
1	124	89	71.8%	89	124
2	132	84	63.6%	173	256
3	118	71	60.2%	244	374

...

12 | 145 | 45 | 31.0% | 1,089 | 3,520 (30.9% cumulative churn)

Narrative: "By month 12, we've lost 30.9% of customers cumulatively. The churn rate **drops sharply** after month 6 (from 60%+ to <35%), indicating successful onboarding by month 6. Focus: reduce 0–6 month loss from 60% to 40% = recover ~60 customers/month = \$720K/year."

SECTION 11: Composite Insights (Multi-Factor Analysis)

Worst Case Scenario:

Criteria:

- Month-to-month contract
- Tenure $\leq$ 12 months
- No security/tech support
- Monthly charge >\$50

Results:

- Size: 523 customers
- Churn Rate: 76.1%
- Monthly Revenue at Risk: \$34.2K

Best Case Scenario:

Criteria:

- 1-year or 2-year contract
- Security + Tech Support + Backup
- Any tenure

Results:

- Churn Rate: 2.1%
- Avg Tenure: 51 months

Business Narrative: "The contrast is stark. Our '**perfect storm**' customers (MTM + new + no add-ons) churn at 76%, while '**ideal**' customers (locked-in + bundled) churn at 2%. The path is clear: move customers from storm to ideal through contract upgrades and cross-sells."

SECTION 12: Actionable Metrics (Strategic Levers)

Lever #1: Contract Upgrade

Month-to-month customers (No churn yet): 2,200

Avg monthly revenue: \$68.50

Total monthly at risk: \$150.7K

Goal: Move 15% to 1-year contracts

Incentive cost: \$5/month discount = \$16.5K/month

Expected churn reduction: from 42.7% → 18% = 24.7% improvement = 544 customers saved/month

ROI: $(\$544 \times \$68.50 \times 12 \text{ months}) - (\text{upgrade incentive} \times 330 \text{ customers} \times 12 \text{ months}) = \447K/year

Lever #2: Service Bundle Cross-Sell

Customers with no add-ons (and not churned): 3,200

Current churn rate: 26.5%

If we move to 2+ services: 9.1% churn

Gap: 17.4 percentage points = 557 customers

Value: $557 \times \$68/\text{month} \times 12 = \454K/year

Cost: Discounted bundle (tech support + security = \$15/month offered at \$8) = \$2.2M/year for all

Net ROI: 20% adoption = 640 customers moved, churn drop = 112 saved = \$91K/year

Still positive if we achieve 30%+ adoption.

Lever #3: Fiber Expansion (Infrastructure)

DSL customers: 2,400

DSL churn: 25.8%

Fiber churn: 11.6%

Churn difference: 14.2 percentage points = 341 customers/month

If we can convert 20% of DSL to fiber in 12 months:

- 480 customers upgraded

- Churn reduction: $480 \times 14.2\% = 68$ customers saved/month

- Annual impact: $816 \text{ customers} \times \$47.50/\text{month} = \$464\text{K/year}$ (assuming avg DSL charge maintained)

CapEx: $\sim \$2\text{K per fiber installation} \times 480 = \960K (breakeven in 2 years with churn savings alone)



Quick Reference: Impact Estimates

Lever	Action	Effort	Impact	Annual Revenue
Contract Upgrade	Incentivize 1-year upgrade	Medium	15% of MTM segment	\$447K
Service Bundle	Cross-sell Tech Support + Security	Low	20% adoption → 3% churn drop	\$91K+
Onboarding	Proactive support calls weeks 2,4,6	Medium	10–12% reduction in 0–6mo churn	\$600K
Fiber Expansion	Upgrade DSL → Fiber (20% cohort)	High CapEx	14.2% churn gap closed	\$464K
Risk Segment	Target 234 CRITICAL customers	Low	40% retention = \$38K/year	\$38K
TOTAL	Bundled strategy	Medium	5–8% company churn reduction	\$1.6M+

— END OF ANALYSIS —